Start here | Priority Pre emptive.c

```c
#include <stdio.h>

int main()
{
    int n, choice;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], pr[n], rt[n];
    int ct[n], tat[n], wt[n];
    int pid[n];

    printf("Enter priorities (1 for higher number means higher priority, 2 for lower number means higher priority): ");
    scanf("%d", &choice);

    printf("Enter priorities:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&pr[i]);

    printf("Enter arrival times:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&at[i]);

    printf("Enter burst times:\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&bt[i]);
        rt[i] = bt[i];
        pid[i] = i+1;
    }

    int time = 0, completed = 0;

    while(completed < n)
    {
        int index = -1;

        for(int i=0;i<n;i++)
        {
            if(at[i] <= time && rt[i] > 0)
            {
                if(index == -1)
                    index = i;
                else
                {
                    if(choice == 1)
                    {
```

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```
                    {
                        if(pr[i] > pr[index])
                            index = i;
                    }
                    else
                    {
                        if(pr[i] < pr[index])
                            index = i;
                    }
                }
            }
        }

        if(index != -1)
        {
            rt[index]--;
            time++;

            if(rt[index] == 0)
            {
                ct[index] = time;
                tat[index] = ct[index] - at[index];
                wt[index] = tat[index] - bt[index];
                completed++;
            }
        }
        else
        {
            time++;
        }
    }

    float avg_tat = 0, avg_wt = 0;

    printf("\nPriority scheduling (Pre-Emptive):\n");
    printf("PID\tPrior\tAT\tBT\tCT\tTAT\tWT\tRT\n");

    for(int i=0;i<n;i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
               pid[i], pr[i], at[i], bt[i], ct[i], tat[i], wt[i], rt[i]);

        avg_tat += tat[i];
        avg_wt += wt[i];
    }
```

```
    }

    avg_tat /= n;
    avg_wt /= n;

    printf("\nAverage turnaround time: %.2f\n", avg_tat);
    printf("Average waiting time: %.2f\n", avg_wt);

    return 0;
}
```

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```
"D:\1WA24CS231\Priority Pre

Enter number of processes: 7
Enter priorities (1 for higher number means higher priority, 2 for lower number means higher priority): 2
Enter priorities:
3 4 4 5 2 6 1
Enter arrival times:
0 1 3 4 5 6 10
Enter burst times:
8 2 4 1 6 5 1

Priority scheduling (Pre-Emptive):
PID     Prior   AT      BT      CT      TAT     WT      RT
P1      3       0       8       15      15      7       0
P2      4       1       2       17      16      14      0
P3      4       3       4       21      18      14      0
P4      5       4       1       22      18      17      0
P5      2       5       6       12      7       1       0
P6      6       6       5       27      21      16      0
P7      1       10      1       11      1       0       0

Average turnaround time: 13.71
Average waiting time: 9.86

Process returned 0 (0x0)   execution time : 44.071 s
Press any key to continue.
```

Round Robin

```c
#include <stdio.h>

int main() {

    int n, tq;
    int at[20], bt[20], rt[20];
    int wt[20], tat[20], rt_time[20];
    int time = 0, completed = 0;

    float avg_wt = 0, avg_tat = 0, avg_rt = 0;

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(int i=0;i<n;i++){
        printf("Enter Arrival Time and Burst Time for P%d: ",i+1);
        scanf("%d %d",&at[i],&bt[i]);

        rt[i] = bt[i];
        rt_time[i] = -1;
    }

    printf("Enter Time Quantum: ");
    scanf("%d",&tq);

    while(completed < n){

        int executed = 0;

        for(int i=0;i<n;i++){

            if(at[i] <= time && rt[i] > 0){

                if(rt_time[i] == -1)
                    rt_time[i] = time - at[i];

                if(rt[i] > tq){
                    time += tq;
                    rt[i] -= tq;
                }
                else{
                    time += rt[i];
                    rt[i] = 0;

                    tat[i] = time - at[i];
                    wt[i] = tat[i] - bt[i];
```

```
                    avg_wt += wt[i];
                    avg_tat += tat[i];
                    avg_rt += rt_time[i];

                    completed++;
                }

                executed = 1;
            }
        }

        if(executed == 0)
            time++;
    }

    printf("\nPID\tAT\tBT\tWT\tTAT\tRT\n");

    for(int i=0;i<n;i++){
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,at[i],bt[i],wt[i],tat[i],rt_time[i]);
    }

    printf("\nAverage Waiting Time = %.2f",avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f",avg_tat/n);
    printf("\nAverage Response Time = %.2f\n",avg_rt/n);

    return 0;
}
```

```
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Enter number of processes: 5
Enter Arrival Time and Burst Time for P1: 0 5
Enter Arrival Time and Burst Time for P2: 1 3
Enter Arrival Time and Burst Time for P3: 2 1
Enter Arrival Time and Burst Time for P4: 3 2
Enter Arrival Time and Burst Time for P5: 4 3
Enter Time Quantum: 2

PID     AT      BT      WT      TAT     RT
P1      0       5       9       14      0
P2      1       3       8       11      1
P3      2       1       2       3       2
P4      3       2       2       4       2
P5      4       3       6       9       3

Average Waiting Time = 5.40
Average Turnaround Time = 8.20
Average Response Time = 1.60

Process returned 0 (0x0)   execution time : 86.091 s
Press any key to continue.
```

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Priority non pre emptive

```c
#include <stdio.h>


int main()

{

    int n, i, current_time = 0, completed_count = 0;


    printf("Enter number of processes: ");

    scanf("%d", &n);


    int pid[n], at[n], bt[n], pr[n];

    int ct[n], tat[n], wt[n], rt[n], st[n];

    int completed[n];


    // Input

    for(i = 0; i < n; i++)

    {

        printf("\nEnter Process ID: ");

        scanf("%d", &pid[i]);


        printf("Enter Arrival Time: ");

        scanf("%d", &at[i]);


        printf("Enter Burst Time: ");

        scanf("%d", &bt[i]);


        printf("Enter Priority: ");

        scanf("%d", &pr[i]);


        completed[i] = 0;

```

```
    }


    while(completed_count < n)

    {

        int highest_priority = 9999;

        int selected = -1;


        for(i = 0; i < n; i++)

        {

            if(at[i] <= current_time && completed[i] == 0)

            {

                if(pr[i] < highest_priority)

                {

                    highest_priority = pr[i];

                    selected = i;

                }

            }

        }


        if(selected == -1)

        {

            current_time++;

        }

        else

        {

            st[selected] = current_time;

```

```
            ct[selected] = st[selected] + bt[selected];

            tat[selected] = ct[selected] - at[selected];

            wt[selected] = tat[selected] - bt[selected];

            rt[selected] = st[selected] - at[selected];

            current_time = ct[selected];

            completed[selected] = 1;

            completed_count++;

        }

    }

    printf("\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\tRT\n");

    for(i = 0; i < n; i++)

    {

        printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\t%d\n",

        pid[i], at[i], bt[i], pr[i], ct[i], tat[i], wt[i], rt[i]);

    }

    return 0;

}
```

```
"D:\1WA24CS231\Priority Non

Enter Process ID: 1
Enter Arrival Time: 0
Enter Burst Time: 3
Enter Priority: 5

Enter Process ID: 2
Enter Arrival Time: 2
Enter Burst Time: 2
Enter Priority: 3

Enter Process ID: 3
Enter Arrival Time: 3
Enter Burst Time: 5
Enter Priority: 2

Enter Process ID: 4
Enter Arrival Time: 4
Enter Burst Time: 4
Enter Priority: 4

PID     AT      BT      PR      CT      TAT     WT      RT
1       0       3       5       3       3       0       0
2       2       2       3       10      8       6       6
3       3       5       2       8       5       0       0
4       4       4       4       14      10      6       6

Process returned 0 (0x0)   execution time : 67.597 s
Press any key to continue.
```

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